

HiddenBench Imitation Control Study

Validation agenda, sanity checks, and weekly program

Working research note – 10 August 2026

1. What is now fixed

The current scientific object is a **reasoning-augmented multi-opinion imitation process with feedback control**. It deliberately combines two established objects rather than inventing an unconstrained new game. HiddenBench supplies the semantic task: shared information, distributed private evidence, a finite answer set, and a known correct answer. The social-imitation model supplies the dynamical skeleton: a population occupies a finite set of opinions and evolves through one-agent opinion changes.

For one HiddenBench task, let

$$\mathcal{A} = \{a_1, \dots, a_K\}, \quad Y^* \in \mathcal{A}$$

be the answer set and ground-truth answer. Agent i receives local evidence E_i and has a committed opinion

$$X_i(t) \in \mathcal{A}.$$

The microscopic state and occupation state are

$$X_t = (X_1(t), \dots, X_N(t)), \quad \mathbf{n}_t = (n_1(t), \dots, n_K(t)),$$

with $n_a(t) = \sum_i \mathbf{1}[X_i(t) = a]$ and $p_a(t) = n_a(t)/N$. One elementary event changes at most one focal agent,

$$X_i(t) = a \longrightarrow X_i(t+1) = b, \quad \mathbf{n}_{t+1} = \mathbf{n}_t - \mathbf{e}_a + \mathbf{e}_b.$$

This one-step geometry is the bridge to the Irisarri social-imitation dynamics.

The central experimental switch is

reasoning ON: LLM update using HiddenBench evidence and interaction history,
reasoning OFF: provider-free classical multi-opinion imitation kernel.

Everything else should be matched as closely as possible: task, answer alphabet, population size, initial vote vector, interaction schedule, controller target, controller observation rule, and logging. This is the crucial comparison. The question is not merely whether an LLM population can be controlled; it is *what changes in the controlled dynamics when semantic reasoning replaces a classical imitation kernel*.

Order parameters and observables

The first observables are deliberately simple. Let $Z \in \mathcal{A}$ be the controller target. Define

$$m_t^{\text{truth}} = \frac{Kp_{Y^*}(t) - 1}{K - 1}, \quad m_t^{\text{ctrl}} = \frac{Kp_Z(t) - 1}{K - 1},$$
$$m_t^{\text{order}} = \frac{K \max_a p_a(t) - 1}{K - 1}, \quad \tilde{H}_t = -\frac{1}{\log K} \sum_a p_a(t) \log p_a(t).$$

The first measures alignment with truth, the second alignment with the controller, the third ordering irrespective of which option wins, and the fourth population-level disorder. Evidence disclosure and evidence reach remain separate observables because HiddenBench's core failure is information surfacing rather than inability to reason once information is available.

2. The controller: what is stochastic and what is not

The controller should be understood as a measurement-feedback system,

$$X_t \longrightarrow Y_t \longrightarrow U_t \longrightarrow X_{t+1}.$$

This is the part that should remain close in spirit to Horowitz–Sandberg. The population is the system. The controller does not observe the system perfectly; it receives a stochastic measurement Y_t . It then chooses a control action U_t , which locally modifies the next interaction. The essential randomness is therefore not an arbitrary coin toss added merely to make $H(U_t) > 0$.

For the pilot, the cleanest sensor is finite population sampling. If the controller samples q_c agents at event t , then Y_t is their observed opinion-count vector. Conditional on the occupation state, Y_t follows the appropriate finite-population sampling distribution (hypergeometric without replacement, or multinomial if sampling with replacement). Thus Y_t is random even when $\mathbf{n}_t$ is fixed. The quantity q_c has a clear interpretation: it controls **measurement quality**. Small q_c gives a noisy view of the population; large q_c gives a more accurate view.

The pilot policy may be deterministic,

$$U_t = \pi_C(Y_t),$$

for example $U_t \in \{\text{NO_OP}, \text{ADVOCATE_Z}\}$ with an intervention when sampled support for Z is below a threshold. Because Y_t is stochastic, U_t will generally vary across trajectories even if π_C itself is deterministic. A stochastic policy $\pi_C(u \mid Y_{\leq t})$ can be introduced later, but it is not required merely to obtain nonzero controller entropy.

When $U_t = \text{ADVOCATE_Z}$, control is **local**: the focal population agent receives a controller interaction advocating Z . In reasoning mode, the controller must not overwrite the vote; the LLM may accept or reject the intervention. The initial controller should not possess HiddenBench private evidence or the task rationale. In classical mode, the same event should enter as an explicit provider-free control channel modifying the transition kernel toward Z .

There are then two distinct information directions:

$$\boxed{\text{system} \rightarrow \text{controller}} \quad \text{and} \quad \boxed{\text{controller} \rightarrow \text{system}}.$$

The first measures sensing/information acquisition. A history-aware form is

$$\mathcal{T}_{X \rightarrow Y}(t) = I(X_{\leq t}; Y_t \mid Y_{< t}).$$

For the first discrete pilot, a simpler mesoscopic diagnostic such as $I(\mathbf{n}_t; Y_t)$ is useful because the measurement is generated directly from the occupation state. The actuation direction can be probed through

$$\mathcal{T}_{U \rightarrow X}(t) = I(U_t; X_{t+1} \mid X_t),$$

or, to reduce sparsity in the first count-based analysis,

$$I(U_t; X_{i_t}(t+1) \mid X_{i_t}(t), \mathbf{n}_t).$$

The latter is especially attractive at small N : it conditions on the focal opinion and mesoscopic state rather than the entire microscopic configuration.

3. Sanity checks immediately after implementation

The first run should be treated as a scientific unit test, not as a result. The following checks should be passed before spending on a real grid.

A. Dynamical invariants

- Exactly one focal opinion may change per elementary event; all non-focal opinions remain unchanged.
- $\sum_a n_a(t) = N$ for every event and the logged occupation vector must reconstruct the logged individual vote vector.
- The raw time index and normalized population time $\tau = t/N$ must be reproducible from the logs.
- Consensus stopping should be disabled for the first information experiments so matched trajectories have equal horizons.

B. HiddenBench semantics

- No agent receives hidden evidence it was not assigned, and the controller never receives hidden facts, rationale, or the answer explanation.
- The initial Hidden Profile vote should remain difficult while Full Profile remains substantially easier. Because the new interaction protocol differs from the original HiddenBench paper, the strongest direct comparison is the *pre-interaction* vote, not post-discussion performance.
- Option labels should be permuted in at least one sanity run. The raw HiddenBench corpus is position-imbalanced, so a controller effect must not be an answer-position artifact.

C. Reasoning OFF really recovers the classical process

- Classical mode makes zero provider calls.
- Given the same X_0 , the classical run evolves exclusively through explicit one-agent $a \rightarrow b$ jumps generated by the configured imitation rates/weights.
- Forward/reverse reaction channels configured as microscopically reversible must both have support. Log reaction identity and transition weight/rate for every jump.
- Run a large number of cheap classical trajectories first. Their empirical transition frequencies should agree with the implemented transition probabilities/rates within Monte Carlo error.

D. Controller mechanics

- With control disabled, the controller path is truly absent and the game reproduces the no-controller dynamics under matched seeds/configuration.
- Y_t must be sampled from the *current* population. For a hand-constructed occupation vector, the empirical distribution of sensor counts across repeated samples should match the theoretical hypergeometric/multinomial distribution.
- The policy must produce more than one action in realistic repeated trajectories. If U_t is almost always NO_OP or almost always ADVOCATE_Z, controller-to-system MI will be statistically uninformative even if the code is correct.
- The controller must influence rather than force the focal decision in reasoning mode.
- Perform a target-reversal check: choose a wrong target $Z \neq Y^*$. A genuine controller should be able to increase m^{ctrl} even when this decreases m^{truth} . If the system always moves toward truth regardless of Z , the apparent “control” may actually be a task-solving or prompt artifact.

E. Order-parameter checks

For hand-created states, verify the formulas exactly. For example, uniform occupation over K opinions gives $m^{\text{truth}} = m^{\text{ctrl}} = 0$ when the corresponding option has share $1/K$, and $m^{\text{order}} = 0$. Consensus on any option gives $m^{\text{order}} = 1$. Consensus on truth gives $m^{\text{truth}} = 1$; consensus on the controller target gives $m^{\text{ctrl}} = 1$.

4. What to inspect in the first real trajectories

For one or two tasks, plot the following against interaction index or $\tau = t/N$:

- all per-option shares $p_a(t)$;
- $m^{\text{truth}}(t)$, $m^{\text{ctrl}}(t)$, $m^{\text{order}}(t)$, and $\tilde{H}(t)$;
- evidence disclosure/reach;
- the sensor observation Y_t and controller action U_t as event markers;
- the focal transition $a \rightarrow b$ at each event.

The most useful first summaries are response differences relative to no control,

$$\begin{aligned}\Delta m^{\text{ctrl}}(T) &= \mathbb{E}[m_T^{\text{ctrl}} \mid \text{control}] - \mathbb{E}[m_T^{\text{ctrl}} \mid \text{no control}], \\ \Delta m^{\text{truth}}(T) &= \mathbb{E}[m_T^{\text{truth}} \mid \text{control}] - \mathbb{E}[m_T^{\text{truth}} \mid \text{no control}].\end{aligned}$$

Also inspect empirical transition matrices separately for reasoning ON and OFF. The strongest early qualitative result would be that the same X_0 and controller produce visibly different transition statistics when semantic reasoning is enabled.

For the information analysis, inspect the **support of the contingency tables before trusting the MI value**: counts per $(\mathbf{n}_t, Y_t)$ cell, counts per $(X_{i_t}(t), \mathbf{n}_t, U_t)$ cell, and the empirical entropy $H(U_t)$. Small populations help here because the mesoscopic state space is small. For example, with $K = 3, N = 4$ there are only $\binom{N+K-1}{K-1} = 15$ occupation states. This is why $N = 4\text{--}5$ is perfectly reasonable for the first existence test, even though it is not a regime for thermodynamic-limit or phase-transition claims.

5. What would count as a positive signal?

The agenda is validated progressively. No single plot is sufficient.

1. **Behavioral response.** The controller changes the population relative to no control without directly forcing votes. Ideally there is a reproducible shift in m^{ctrl} and/or transition frequencies.
2. **Sensing information.** $X \rightarrow Y$ information is nonzero and disappears under a measurement-shuffle null. As a basic calibration, it should respond sensibly to sensor quality q_c .
3. **Actuation information.** $U \rightarrow X$ conditional information is nonzero, exceeds shuffled/circular-shift nulls, and has stable bootstrap uncertainty.
4. **Reasoning matters.** Under matched X_0 and controller conditions, reasoning ON and OFF have different response curves, transition kernels, or information-response relationships. This is the key scientific signal. If they are indistinguishable, the HiddenBench semantic layer is not yet doing useful dynamical work.
5. **Target specificity.** Changing Z changes the direction of the response. This distinguishes control from a generic “helpful prompt” effect.

A particularly promising eventual quantity is *response per bit*, but it should only be introduced after the information signals are reliable. At that stage one can compare, for example,

$$\eta_{\text{info}} \sim \frac{\Delta m^{\text{ctrl}}}{\sum_t \mathcal{T}_{U \rightarrow X}(t)},$$

while explicitly calling it information-normalized control efficacy rather than thermodynamic efficiency.

6. Parameters: what is measured versus what is varied

It is useful to separate **order parameters** from **control/system parameters**. The order parameters above are outcomes. They are not the axes of the experiment.

Category	Parameter	Interpretation / priority
Reasoning	ON / OFF	Primary scientific comparison; vary immediately.
Population	N	Finite-size/system-scale axis; keep small this week, scale later.
Imitation	q	Number of social sources influencing a focal agent in the classical model; fix at the simplest value initially, later useful for connection to q-voter theory.
Controller sensor	q_c	Number of population agents observed by the controller; clear measurement-quality axis.
Controller policy	threshold θ_c	Intervention aggressiveness; higher/lower threshold changes how often feedback acts. Keep fixed in the first pilot, sweep only after a signal exists.
Controller target	Z	Correct versus explicit wrong target; use as a specificity/sanity check immediately.
Evidence structure	redundancy / assignment scheme	Important later for population scaling; not a Monday–Tuesday axis.

The first plausible two-dimensional “control phase diagram” is therefore not an order-parameter plot. It would be something like measurement quality versus feedback aggressiveness, (q_c, θ_c) , with color showing m^{ctrl} , m^{truth} , or information-normalized response. Population size N then becomes a finite-size extension. Another later axis is the imitation interaction size q , which is physically appealing because it connects directly to the classical nonlinear voter structure. None of these sweeps is required to validate the agenda this week.

7. Weekly program from the current point

Monday + Tuesday – finish the scientific object and validate it. The implementation agent is currently building the game. Once it runs, perform the sanity checks above. Run cheap/mock/classical trajectories before real LLM calls. Then run a tiny real-provider comparison: reasoning ON/OFF, controller OFF/ON, and one correct versus one wrong target on one or two tasks. Do not optimize the controller yet.

Wednesday – ask only whether the two information directions are statistically measurable. Adapt the existing direct-counting MI/conditional-MI pipeline to the event table. Begin with low-dimensional quantities such

as $I(\mathbf{n}_t; Y_t)$ and $I(U_t; X_{i_t}(t+1) \mid X_{i_t}(t), \mathbf{n}_t)$. Validate first on synthetic/classical data where the coupling is known. Then use bootstrap intervals, permutation nulls, circular-shift nulls, and label-swap checks. Produce sample-size curves. The output of Wednesday should be a statement about estimator feasibility, not a paper claim.

Thursday – first scientific pilot. Run enough repeated LLM episodes to obtain trajectory averages and the two directional information estimates for a small set of validated tasks. Compare matched reasoning ON/OFF runs. Plot controller response, truth response, evidence diffusion, and information flow together. If possible, include the target-reversal control.

Friday – go/no-go decision. Use the following diagnosis:

Observation	Interpretation / next move
Behavioral response + both information directions + ON/OFF difference	Strong green light: freeze the game and scale experiments.
Behavioral response but MI unstable	Statistical/estimator problem: increase repetitions or simplify conditioning before changing the game.
Nonzero $X \rightarrow Y$ but no $U \rightarrow X$ and no response	Sensor works, actuator/controller policy is ineffective; redesign control, not the estimator.
Nonzero $U \rightarrow X$ but no target-specific response	Possible correlation/confounding; inspect policy/state dependence and nulls.
No response and no information signal	The current control mechanism is not a useful probe; change the controller/game interaction before any InfoNCE work.
Reasoning ON and OFF are indistinguishable	The semantic layer is not yet producing a distinct dynamical phenomenon; inspect evidence transmission/update rules before scaling.

8. Post-week extensions, in priority order

If Friday is positive, the next extension should be **scale and structure**, not neural MI estimation immediately. First increase N using the already available exact-replication population construction, for example $N = 8, 16, 32$, while plotting time in units $\tau = t/N$. This establishes whether the phenomenon survives beyond the canonical four-agent Hidden Profile setting. After that, vary one controller/system axis at a time: q_c , controller threshold/aggressiveness, and later imitation interaction size q . Only once the low-dimensional directed-information signal is robust should the project move to history-aware semantic estimation with embeddings/InfoNCE.

The stochastic-thermodynamic extension comes later still. Irisarri et al. obtain fluctuation relations by using microscopically reversible reaction channels and explicit forward/backward trajectory probabilities. The classical mode is constructed to preserve that possibility; the LLM mode should not be assumed to satisfy it. A controlled second-law-like statement is therefore a later theory objective, not a requirement for the current empirical agenda.

Bottom line

The immediate research question is now narrow:

Does stochastic feedback produce a measurable, target-specific control signal in HiddenBench imitation dynamics, and does semantic reasoning change that signal?

Small N is sufficient for this first existence and estimator-feasibility test because repeated trajectories provide the statistical sample and the mesoscopic state space remains compact. Large N becomes necessary only when the project moves from “does the phenomenon exist?” to finite-size scaling, critical behavior, and population-level control regimes.

Anchoring sources. Y. Li, A. Naito, H. Shirado, “Systematic Failures in Collective Reasoning under Distributed Information in Multi-Agent LLMs,” ICML 2026. L. Irisarri, L. Trigo, R. Toral, G. Manzano, “Stochastic Thermodynamics of Social Imitation beyond Energetics,” 2026. J. M. Horowitz, H. Sandberg, “Second-law-like inequalities with information and their interpretations,” 2014. Current repository implementation plan and architecture notes for `hidden_bench_imitation`.